

# Inhibitory effect of aqueous extracts of some herbs on human platelet aggregation in vitro.

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Effect of aqueous extract of several herbs on human platelet aggregation in vitro was investigated. Out of 28 herbs/nutraceuticals investigated, camomile, nettle alfalfa, garlic and onion exhibited most significant anti-platelet activity ( $\geq 45\%$  inhibition). Aqueous extracts of alfalfa, fresh nettle, and camomile inhibited ADP induced-platelet aggregation by 73, 65 and 60%, respectively, compared with control ( $P < 0.05$ ). Camomile and alfalfa inhibited collagen-induced platelet aggregation by 84 and 65%, respectively, but nettle could not inhibit collagen-induced aggregation. In contrast, nettle was the most potent inhibitor (66%) of whole blood aggregation induced by collagen, followed by alfalfa (52%), and camomile (30%) compared with control ( $P < 0.05$ ). None of these three herbs however could inhibit arachidonic acid or thrombin induced platelet aggregation. Camomile and alfalfa strongly inhibited thromboxane B<sub>2</sub> synthesis induced by ADP or collagen, but nettle had no effect. Alfalfa and nettle increased cGMP levels in platelets by 50 and 35%, respectively, compared with the control ( $1.85 \pm 0.23$  nM) ( $P < 0.005$ ). All these data indicate that camomile, nettle and alfalfa have potent anti-platelet properties, and their inhibitory actions are mediated via different mechanisms.